Scout Troop 1018 Merit Badge Opportunities

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I am pleased to offer merit badge counseling for several badges related to technology, engineering, and space science. This document summarizes the merit badges currently offered, provides a brief overview of their topics and representative requirements, and outlines the procedures for starting and completing a merit badge with the counselor.

The most current merit badge requirements appear in the *Scouts BSA Requirements* book and on *Scouting America Merit Badge Hub*. Any Scout beginning work on a merit badge must follow the requirements exactly as stated in those official sources.

1 Merit Badges Offered

The following summaries describe the overall topic and representative requirements for the seven merit badges for which I am currently certified as a counselor.

Artificial Intelligence

Overall topic: Foundations of AI, real-world uses, ethics, and responsible decision-making.

Key requirements:

- Explain core AI ideas, including distinguishing AI from simple automation.
- Discuss real-world applications of AI and how AI systems use data.
- Explore ethical issues such as bias, privacy, accountability, and impacts on society.
- Complete an AI-related practical activity or project approved by the counselor.
- Investigate careers related to AI and the preparation needed for those fields.

Official document: artificial-intelligence

Programming

Overall topic: Programming languages, software concepts, intellectual property, and writing/debugging code.

Key requirements:

- Review digital safety and discuss basic health and safety issues related to programming.
- Discuss the history of programming and major milestones in its development.
- Compare commonly used programming languages and devices that rely on programs.
- Learn basic software intellectual property concepts such as licensing and open source.
- Write, debug, demonstrate, and explain programs in multiple programming languages or environments.

Official document: Programming.pdf

Robotics

Overall topic: Robot systems, sensors, programming, design process, and demonstration of a working robot.

Key requirements:

- Explain robotics safety, hazards, and first aid considerations.
- Discuss what robots do and compare remote-control, telerobotic, and autonomous systems.
- Learn major robotics fields such as mobility, manipulation, sensors, and programming.
- Design, build, program, and test a robot or robotic subsystem with sensor feedback.
- Maintain a robot engineering notebook and demonstrate the finished robot.

Official document: Robotics.pdf

Digital Technology

Overall topic: Computers, networks, software, digital media, cybersecurity, projects, and responsible technology use.

Key requirements:

- Review digital safety and discuss how digital technology has changed over time.
- Explain digitization, compression, networks, malware, and secure web connections.
- Complete hands-on digital projects such as spreadsheets, documents, graphics, presentations, blogs, or webpages.
- Discuss intellectual property and responsible use of software and online information.
- Learn about recycling, disposal, and career opportunities in digital technology.

Official document: Digital_Technology.pdf

Engineering

Overall topic: How engineers solve problems, engineering fields, design process, ethics, and careers.

Key requirements:

- Investigate how a manufactured item works and what engineering was needed to create it.
- Research an engineering achievement that had a major impact on society.
- Explain several engineering disciplines and how engineers contribute to projects.
- Meet with an engineer or learn directly about engineering work and tools.
- Use a systems engineering approach to design something and discuss engineering ethics and careers.

Official document: Engineering.pdf

Space Exploration

Overall topic: Purposes of space exploration, rockets, satellites, missions, habitats, and careers in space.

Key requirements:

- Explain the purposes of space exploration, including historical and scientific motivations.
- Build and launch a model rocket, or complete the alternate activity where launching is not allowed.
- Identify and explain the major parts of a rocket.
- Discuss rocket propulsion, action-reaction, orbits, and satellite imaging.
- Explore space missions, design concepts for a space habitat or settlement, and discuss careers.

Official document: Space_Exploration.pdf

Electronics

Overall topic: Circuit basics, schematics, soldering, digital and analog applications, testing, and careers.

Key requirements:

- Explain safety precautions for building, altering, and repairing electronic devices.
- Draw and explain a simple schematic using standard symbols.
- Demonstrate soldering and desoldering and discuss printed circuit board precautions.
- Discuss control, digital logic, and analog applications, then complete an electronics project.
- Apply Ohm's law, use test equipment, read resistor color codes, and explore electronics careers.

Official document: Electronics.pdf

2 Procedure for Starting a Merit Badge

Role of the Merit Badge Counselor. The role of a Merit Badge Counselor is to provide guidance, mentoring, and evaluation as Scouts work through the merit badge requirements. The counselor does not function as a formal instructor. Scouts are expected to review the merit badge materials, take initiative in learning, and demonstrate completion of each requirement.

Steps to Begin

1. **Talk with your Scoutmaster first.** Before starting any merit badge, Scouts must receive approval from the Scoutmaster.
2. **Obtain a Blue Card.** Complete the blue card attached at the end of this document and obtain the Scoutmaster's signature on the front of the card.
3. **Contact the Merit Badge Counselor.** Send an email to the counselor (Taeyoung Lee at tylee.fdcl@gmail.com) including:
   - Scout's name
   - Merit badge requested
   - Parent or guardian copied on the email
4. **Schedule the first meeting.**

   The first meeting will review:
   - Merit badge requirements
   - Project expectations
   - Timeline for completion
5. **Work on requirements.** Scouts should review the merit badge pamphlet and begin completing requirements between meetings.
6. **Completion and Blue Card sign-off.** Once all requirements are completed, the Merit Badge Counselor signs the blue card and records completion.

Important Guidelines

- **Parents or guardians should be copied on all communication.**
- **Buddy system.** In accordance with BSA Youth Protection guidelines, Scouts must not meet one-on-one with an adult. When meeting with the Merit Badge Counselor, a Scout should be accompanied by a buddy (another Scout, a sibling, or a parent/guardian), or the meeting should take place in a group setting.
- Scouts must complete **all requirements exactly as written** in the official merit badge requirements.
- Some badges involve projects, demonstrations, or presentations.
- Scouts may work individually or in small groups depending on the activity.

3 Group Merit Badge Instruction

For certain merit badges with strong interest or collaborative activities, the counselor may organize small group sessions to provide guidance and facilitate discussion. These sessions are intended to support Scouts as they work through the requirements; however, each Scout is responsible for completing all merit badge requirements individually as written. Participation in a group session does not guarantee that a merit badge will be earned; each Scout must still demonstrate completion of every requirement to the satisfaction of the Merit Badge Counselor.

Currently, group sessions are planned for the following two merit badges.

Artificial Intelligence

There is expected to be strong interest in the *Artificial Intelligence* merit badge. To support Scouts working on this badge, the counselor plans to organize a series of **3–4 group sessions** focused on discussion of key concepts and guidance on the requirements.

- **Schedule:** Mondays, 6:30–7:30 PM
- **Tentative timeframe:** April
- **Format:** Interactive discussions and guidance on selected requirements

Additional details regarding the exact schedule, enrollment procedures, and preparation materials will be shared with the troop soon.

Robotics

The *Robotics* merit badge typically involves substantial hands-on experience with robotics systems. Participation in robotics competitions such as *FIRST LEGO League (FLL)* or *VEX IQ Robotics* often provides relevant experience for several of the merit badge requirements.

If any Scout has participated in these or similar robotics programs, please inform the counselor. If sufficient interest exists, a **small group discussion session** may be organized to review how those experiences relate to the Robotics merit badge requirements and to guide Scouts toward completing the remaining requirements.

Information for Applicant

- A merit badge application can be approved only by a registered merit badge counselor.
- You must have a buddy with you (Scout buddy system) at each meeting with the merit badge counselor.
- Turn in your approved application to your unit leader. You will be awarded the merit badge emblem and certificate at a suitable occasion.

Information for Counselor

- Merit badge applications must be signed in advance by the applicant's unit leader.
- The Scout must have his buddy (Scout buddy system) in attendance at all instructional sessions.
- You may not change any requirement, but you may share your knowledge or experience that will make the counseling more interesting and valuable.

Requirement No. and letter	Date of approval	Counselor initial	Requirement No. and letter	Date of approval	Counselor initial

APPLICATION FOR MERIT BADGE

Name _____

Address _____

City _____

is a registered

☐ Boy Scout ☐ Varsity Scout ☐ Venturer

of _____ No. _____
(Troop, team, crew, ship)

District _____

Council _____

and is qualified to begin working for the merit badge noted on the reverse side.

_____ Date

_____ Signature of unit leader

BOY SCOUTS OF AMERICA®

The applicant has personally appeared before me and demonstrated to my satisfaction that he has met all requirements for the (please print)

_____
Merit badge

_____
Name of counselor

_____
Address of counselor

_____
City Zip Code

_____
Telephone number of counselor

_____
Signature of counselor Date

Checked and recorded:

_____
Date Initials

Date certificate and badge presented: _____

Applicant will turn in this portion to his unit leader for record posting.

APPLICANT'S RECORD

Name _____

has given me his completed application for the

_____
Merit badge

Completed on _____/_____/_____ by
Date

_____
Signature of counselor

_____
Signature of unit leader

NOTE TO BOY SCOUT, VARSITY SCOUT, OR VENTURER: Retain this copy for your permanent records.

COUNSELOR'S RECORD

Applicant _____

☐ Troop
☐ Team Unit number _____
☐ Crew

_____
Merit badge

Date completed _____/_____/_____

Remarks:

It is suggested that the counselor keep this record for at least 1 year in case any question is raised later in regard to this award.